Vitrine

Capture-adaptive photo/video to structured, textured, game-ready 3D scenes for cultural-heritage digitisation

Briefing for University of Salford Digital IT • Deployment, security posture and platform justification • 2026-07-02

Executive summary

Vitrine turns ordinary phone photos and video into structured, textured, game-ready 3D scenes for Unreal Engine 5.8 and Blender, for on-premises cultural-heritage digitisation. It ships as a **single hardened Docker image** with **no SaaS dependency**: the web UI binds `127.0.0.1:7860` and is reached only over an SSH tunnel, so captures, splats and meshes never leave institution hardware. An internal agent controller runs *risk-managed* continuous improvement (propose → adversarially verify → commit, with a human owning rebuilds). End-to-end validated 2026-07-02 on 55 hand-held phone frames. Honest maturity: pre-product, single-site R&D; capture quality is the dominant quality bottleneck; per-object segmentation is a known follow-up.

1 What Vitrine is and what it produces

Vitrine is a capture-adaptive pipeline: it diagnoses each input, routes it through photogrammetry, 3D Gaussian Splatting (3DGS) and mesh-extraction stages as the capture quality allows, and delivers a scene as *standard game assets* (baked-PBR FBX/GLB, not a raw splat or point cloud) with explicit `v2g:*` provenance metadata. It vendors the native C++/CUDA 3DGS trainer LichtFeld Studio at a pinned tag (v0.5.3, never forked). The run below is the validated end-to-end path of 2026-07-02: a brass vessel and ketchup-bottle still life shot on a phone.

Stage	Tool / method	Validated output (2026-07-02)
Ingest / decode	DNG/HEIC decode, camera white balance	55/55 hand-held phone frames decoded
SfM registration	COLMAP (ALIKED + LightGlue)	100% frame registration, 10.4k points
3DGS training	LichtFeld Studio v0.5.3 `igs+`, 30k iter	GPU-bound ∼15 min, 4M-Gaussian room splat
Object segmentation	SAM silhouette crop	per-object crop (follow-up area)
Object reconstruction	TRELLIS.2 image → 3D	4096px PBR-textured GLB, 274k verts
Scene assembly	Blender exhibit scene	staged scene carrying `v2g:*` lineage
Web delivery	`<model-viewer>`	in-browser preview
Engine delivery (opt)	FBX game asset, UE 5.8 (Nanite)	textured mesh import, no splat runtime

Delivering meshes rather than splats is deliberate: 3DGS in a game engine is a renderer, not an asset, and carries documented limits (no Lumen GI/reflection interaction, view-dependent appearance baked in, ∼180 MB per 1M splats vs 5–20 MB for a textured mesh, no standard PBR or collision) [13]. Vitrine treats the splat as an intermediate and ships a mesh.

2 The single hardened mono-image and its attack surface

For IT signoff the relevant fact is the small, auditable surface. Everything runs inside one image; there are no ambient external calls, no per-asset cloud storage, and nothing published to the LAN by default. The design is recorded in ADR-022 (secure single-image) and ADR-023 (web consolidation).

Concern	Vitrine posture
Deployment unit	One auditable Docker mono-image; no service sprawl to inventory or patch piecemeal.
Network exposure	Web UI binds `127.0.0.1:7860`; host publish pinned to `127.0.0.1`. The LAN never sees the service.
Remote access	Operator tunnels in: `ssh -N -L 7860:localhost:7860`. No listener on a routable interface.
Third-party tooling	Least-privilege internal venv + OS-user isolation; bundled model/tool code cannot read host or pipeline secrets.
Data egress	None by design. Captures, splats and meshes stay on institution hardware; air-gappable.
Provenance / audit	Every artifact carries `v2g:*` lineage; runs are reproducible from pinned SOTA versions.
Design records	ADR-022 (secure single-image), ADR-023 (web consolidation).

Why this is IT-signable and air-gappable

The default posture is **deny-by-network**: the only way to reach the UI is an authenticated SSH session that the institution already governs, then a local-forward to loopback. No inbound port is open to the LAN or internet, so the pipeline can run fully air-gapped once model weights are staged on disk. Because it is one image with an internal least-privilege venv and OS-user split, the audit question is bounded: one image to scan, one loopback binding to confirm, one provenance scheme to inspect. This directly answers the data-sovereignty and CARE/FAIR requirements that heritage collections impose (§5).

3 Internal agentic resilience: risk-managed continuous improvement

Vitrine embeds an agent controller that diagnoses each capture, selects the pipeline, drives the tools, runs in-flight evaluations, and recovers from failure (restart-resilient model lifecycles, phased VRAM allocation). Improvement is not ad-hoc: the controller fans out sub-agent "meshes" that **propose** changes and **adversarially verify** them before anything is committed — **propose** → **verify** → **commit**, with a human owning the rebuild. Two concrete cases:

- **QE fleet caught a shipping bug pre-commit.** A quality-engineering sub-agent fleet found a dead API-blueprint wiring bug during adversarial verification and blocked it before it reached a build.
- **Self-healing container entrypoint.** When a pinned wheel's ABI drifted under a Torch upgrade, the entrypoint rebuilt the affected CUDA extension from source rather than failing the run, keeping the image reproducible.

This is automated but bounded: agents propose and verify; a human authorises the rebuild that ships. It is a maintenance and reliability property, not autonomy over production.

Residual risks and mitigations

Risk: agentic self-modification could regress the pipeline. **Mitigation:** no change ships without adversarial verification and a human-owned rebuild; nothing auto-deploys. **Risk:** dependency/ABI drift breaks a run. **Mitigation:** exact-pinned SOTA versions plus self-healing rebuild-from-source in the entrypoint. **Risk:** output quality. **Mitigation (and honest limit):** quality is capture-limited, not pipeline-limited — the dominant lever is capture, and per-object silhouette segmentation is a known, tracked follow-up. Vitrine is single-site R&D, not a shipping product.

4 Position versus competitors

No mainstream tool ships the whole capture-to-game-asset path as one sovereign, offline unit. Cloud tools are structurally disqualified for sovereign heritage data; strong desktop tools are single apps, not an integrated pipeline; the closest academic peer stops at splats.

Tool / service	On-prem?	Trains 3DGS?	Per-object?	UE game asset?	Sovereign?
Vitrine	Yes	Yes	Yes (follow-up)	Yes (FBX/N-anite)	Yes
Luma AI	No (cloud)	Yes	No	Splat only	No
Polycam	No (cloud)	Yes	No	Mesh, no scene	No
KIRI Engine	No (cloud)	Yes	Object mesh	No	No
Scaniverse (Niantic)	On-device	Yes	No	No	Partial
RealityScan (Epic)	Yes	No (SfM)	No	Mesh, manual	Yes
Agisoft Metashape	Yes	No (export)	No	Mesh, manual	Yes
Postshot (Jawset)	Yes	Yes	No	Splat only	Yes
Gaussian Heritage [11]	Yes (Docker)	Yes	Yes	Splat only	Yes

The moat is the *combination*, not any single axis: sovereign on-prem delivery, 3DGS training, per-object scene structure, and UE 5.8 game-asset output in one deployable image. The nearest neighbours are RealityScan (on-prem + UE, but no 3DGS training, no automated object scene) [12] and Gaussian Heritage (structuring + Docker, but research-grade splats, no game-asset delivery) [11]; no competitor spans both. Where rivals are genuinely ahead — stated plainly — is capture UX and ergonomics, out-of-box per-object mesh quality, 3DGS trainer polish, and scale/track record. Vitrine's object quality is still capture-limited and iterating.

5 Business case: why on-prem sovereign digitisation

Two decay vectors make structured, open, re-renderable 3D surrogates a preservation asset rather than a novelty.

Physical decay. UNESCO/WRI (July 2025) find **73% of 1,172 non-marine World Heritage sites are highly exposed to water-related hazards, and 21% face multiple overlapping risks** [1]. Catastrophic single-event losses recur: the National Museum of Brazil fire (2018) destroyed most of a ~20-million-item collection, recoverable only through prior imagery [3]; after Notre-Dame (2019), Andrew Tallon's 2010 laser scan (>1 billion points, ~5 mm) became the restoration blueprint [2]. The lesson is consistent: capture beats reconstruction, and the window in which an object still exists and has not yet been captured is closing.

Digital bitrot. Digitising is necessary but not sufficient. Pew (2024) measured that **38% of web pages that existed in 2013 were gone by 2023** [4]; the Digital Preservation Coalition's 2023 "Bit List" reached **87 endangered digital categories** and reclassified unpublished research data as *Critically Endangered* [5]; storage media are not archival (hard drives 3–5 years, LTO tape 10–20 in practice) so only redundancy plus active migration survives [6]. Open, standardised formats age far better — glTF is an ISO/IEC 12113:2022 standard and OpenUSD is vendor-neutral [10]. Vitrine emits a structured, format-migratable asset stack (mesh + baked textures + `v2g:*` lineage) that a future institution can still open; a defunct SaaS account is not.

Why sovereignty, not cloud. Roughly **5–10% of collections are on display**; the rest sit in storage (90–99% by museum type) [7], so any serious programme is high-volume, and recurring per-asset cloud fees scale badly against it. Sacred, rights-encumbered or community-owned material cannot be governed under a third party's terms: CARE principles and Indigenous-data-sovereignty practice require the originating community to control the infrastructure and access, not merely receive a file on someone else's server [9]. FAIR/paradata scholarship treats the digitisation *process* as a research object needing documented provenance — which Vitrine's pinned, containerised, `v2g:*`-tagged runs provide. Demand is documented: the University of Glasgow's £5.6m "Museums in the Metaverse" surveyed 2,000+ people and found **79% interested in digital tools to explore collections not on display** [8].

6 Funding horizon

Vitrine's four-way hook — heritage digitisation, immersive/games-engine delivery (UE 5.8, Nanite), sovereign/on-prem AI, and real games-asset workflow — maps onto a realistic ladder of live opportunities. UK entities are associated to Horizon Europe for 2024–2027 and can bid Cluster 2 directly.

Programme / call	Funder	Fit / positioning
HORIZON-CL2-2027-01-HERITAGE-08 (intangible CH; ∼Sep 2027)	Horizon Europe CL2 [14]	Primary anchor: Vitrine is the technical pipeline WP inside a culture-led consortium.
HORIZON-CL2-2026-01-HERITAGE-03 (AI in CCSI practice; ∼Sep 2026)	Horizon Europe CL2 [14]	Cleaner "efficient AI workflow" fit, one year sooner; data-sovereign case study.
N-RICH Prototype / Towards a National Collection (2025–27)	UKRI / AHRC [15]	Sovereign, cyber-hardened on-prem digitisation capability answers N-RICH frameworks.
CoSTAR Network (Live/National/Realtime Labs)	UKRI / AHRC [16]	Supplies game-ready structured assets for real-time / virtual-production pipelines.
Innovate UK Creative Catalyst (£150–200k)	Innovate UK	DreamLab AI leads, UoS collaborates; productise the capture-to-UE tool.
MITIH (MediaCity Immersive Tech Hub)	Innovate UK / UoS	Immediate, local, low-friction proof point on a real GM collection.
National Lottery Heritage Grants (rolling, ≤£250k)	NLHF	Fund a cultural partner to run a real digitisation case study.

Positioning. Sequence the domestic funders as rungs that de-risk the 2027 EU anchor: MITIH and Innovate UK for near-term productisation and a local proof point; CoSTAR and AHRC/N-RICH to embed Vitrine as shared real-time capability and a national sovereign-digitisation asset; National Lottery Heritage Grants to generate the real-world case histories the Community-of-Practice pilot bid promised. Each rung produces the partnerships and evidence the HERITAGE-08 consortium bid needs, and every rung leads with the same sovereign, own-your-data framing that the hardened mono-image makes credible.

References

[1] UNESCO/WRI — *Nearly three-quarters of World Heritage sites at high risk from water-related hazards* (2025). https://whc.unesco.org/en/news/2788

[2] Tallon Notre-Dame laser scan (2010; >1B points, ∼5 mm; restoration blueprint) — GIM International. https://www.gim-international.com/content/news/the-indispensable-role-of-point-cloud-data-in-rebui

[3] National Museum of Brazil fire (collection scale; 3D reconstruction from prior imagery). https://en.wikipedia.org/wiki/National_Museum_of_Brazil_fire

[4] Pew Research — *When Online Content Disappears* (link rot, 2024). https://www.pewresearch.org/data-labs/2024/05/17/when-online-content-disappears/

[5] Digital Preservation Coalition — *Global Bit List 2023* (87 endangered categories). https://www.dpconline.org/our-work/digitally-endangered-species

[6] CLIR Pub 54 — *Life Expectancy: How Long Will Magnetic Media Last?* https://www.clir.org/pubs/reports/pub54/4life_expectancy/

[7] ICOM / ICCROM — *Collections in Storage* (∼5–10% displayed). https://icom.museum/wp-content/uploads/2024/05/Report_ICOM-STORAGE_EN_Final.pdf

[8] University of Glasgow — *Museums in the Metaverse* (£5.6m; 79% interested), via Blooloop. https://blooloop.com/museum/in-depth/emerging-museum-technologies/
[9] Indigenous data sovereignty / CARE and digital repatriation. https://terentia.io/blog/indigenous-data-soverei ; https://en.wikipedia.org/wiki/Digital_repatriation
[10] Khronos — OpenUSD/glTF interoperability (glTF = ISO/IEC 12113:2022). https://www.khronos.org/blog/building-bridges-in-3d-aousd-and-khronos-collaborate-on-openusd-and-gltf-interoperability
[11] Gaussian Heritage (3DGS + integrated per-object segmentation; Docker), arXiv 2409.19039. https://arxiv.org/abs/2409.19039
[12] RealityScan 2.1 (on-prem photogrammetry; COLMAP/RF export; does not train 3DGS) — CG Channel. https://www.cgchannel.com/2025/11/epic-games-releases-realityscan-2-1/
[13] 3DGS-in-engine limitations and the mesh path. https://www.polyvia3d.com/guides/gaussian-splatting-unity-un ; https://www.strayspark.studio/blog/gaussian-splatting-unreal-engine-5-capture-to-game-pipeline
[14] Horizon Europe Work Programme 2026–2027, Cluster 2 (HERITAGE topics). https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2026-2027/wp-5-culture-creativit horizon-2026-2027_en.pdf
[15] N-RICH Prototype / Towards a National Collection (UKRI/AHRC). https://www.nationalcollection.org.uk/n-rich-prototype
[16] CoSTAR — Convergent Screen Technologies and Performance in Realtime (UKRI/AHRC). https://www.ukri.org/councils/ahrc/remit-programmes-and-priorities/convergent-screen-technologies-and-performa